Summatory Counting Functions, The DFT, Logic Gates, & Explicit Floor-Free Filtering Schemes for Primes

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1 Building Blocks

The following explicit representation of the divisor-indicator function has been established by several authors, returning 1 if n divides N and 0 otherwise.

$$\hat{D}\left(N,n\right)=\frac{1}{n}\sum_{k=1}^{n}\cos\left(2\pi\frac{kN}{n}\right)=\mathbb{1}_{n|N} \tag{1}$$

This expression has always fascinated me, no matter how many times having proved it through a geometric series, as it seems to me to be too simple to bridge the gap between discrete number theory and standard algebra.

For those interested in the number theoretic applications of this formula, particularly in the context of Mobius inversion, see *Jose Sousa - An Exact Formula for the Prime Counting Function* for marvelous work on the subject.

One can count the factors of n through a sum: $\sum_{j=2}^{n}\hat{D}\left(n,j\right)$, and if this value is 1, then n must be prime. Further, only if this sum is 1 would it divide into 1, thus providing an immediate floor-free indicator, $\hat{p}$, for if a number is prime.

$$\hat{p}\left(n\right)=\hat{D}\left(1,\sum_{j=2}^{n}\hat{D}\left(n,j\right)\right)=\hat{D}\left(2,\sum_{j=1}^{n}\hat{D}\left(n,j\right)\right) \tag{2}$$

2 Logic & Filtering With Indicators

Here, $\hat{\kappa}$ will denote the indicator function corresponding to the boolean true/false statement κ evaluated at n;

$$\hat{\kappa}(n)=\mathbb{1}_{\kappa}=\begin{cases}1, & \text{if } \kappa \text{ is true}\\ 0, & \text{otherwise}\end{cases} \tag{3}$$

From here, computational 'logic' can be applied, such as;

$$\hat{\alpha}\text{ and }\hat{\beta}=\hat{\alpha}\hat{\beta}\ ,\ \text{not }\hat{\kappa}=1-\hat{\kappa}$$

As trivial as these ideas are, the simple combination of indicators together with logic can produce an array of intriguing results. Equipped with the exact numerical formulas from Section 1, one can begin combining as desired.

2.1 Examples

1. Establish an indicator.

    - A twin prime, t exists when both t and $t+2$ are prime. So, $\hat{t}(n)=\hat{p}(n)\hat{p}(n+2)$

2. To form a *counting* function, simply sum over all possible inputs.
   - $\tau(x) = \sum_{n \leq x} \hat{t}(n)$
3. To *evaluate* an expression on values which pass the indicator, simply multiply.
   - The Dirichlet convolution of two arithmetic functions f and g requires summing over all pairs of divisors of n, evaluating at each. $f \star g = \sum_{ab=n} f(a) g(b) = \sum_{k=1}^{n} \hat{D}(n,k) f(k) g\left(\frac{n}{k}\right)$
4. And the possibilities are endless!

Concocting bizarre identities can be done quite swiftly from these above principles;

$$Cop\hat{r}ime(A,B) = D\left(1, \sum_{k=1}^{A} D(A,k) D(B,k)\right) \quad A \geq B$$

$$\mu(n) = \sum_{k=1}^{n} \cos\left(\frac{2\pi k}{n}\right) Cop\hat{r}ime(k,n) \quad \text{The Mobius Function}$$

3 Inequalities With Dirichlet Series

Perhaps the most strange usage of these exact indicators is in the context of Dirichlet-type summations: (Where κ becomes one of the indicators previously discussed.)

$$f_N(s) = \sum_{n=1 \ or \ 2}^{N} \frac{\kappa(n)}{n^s}$$

Using floor $\left(\frac{1}{\sum_{j=2}^{n} D(n,j)}\right)$ in place of $\hat{p}(n)$, prime-related Dirichlet series can be bounded by peculiar reciprocal sums involving the divisor-count function $\sigma(n) =$ *number of factors of n that are not equal to 1.* Letting:

$$P_N(s) = \sum_{p \leq N} \frac{1}{p^s} = \sum_{n=2}^{N} \frac{\text{floor}\left(\frac{1}{\sigma(n)}\right)}{n^s}$$

We observe that:

$$\sum_{n=2}^{N} \frac{\frac{1}{\sigma(n)} - 1}{n} < P_N(s) \leq \sum_{n=2}^{N} \frac{\frac{1}{\sigma(n)}}{n} \Rightarrow 0 < \sum_{n=2}^{N} \frac{1}{n^s \sigma(n)} - P_N(s) \leq \sum_{n=2}^{N} \frac{1}{n^s} \tag{4}$$

Crucially it is the strict lower *inequality* of $x - 1 < \text{floor}(x) \leq x$ which is most helpful. Fixing $s > 1$ and taking N to infinity produces a weak bound on $\gamma(s) = \sum_{n=2}^{\infty} \frac{1}{\sigma(n) n^s}$ between the Prime-Zeta function, $P(s)$ and $\zeta(s) + P(s)$. More interestingly, by setting $s = 1$ or $s = 0$ and using N as a variable, the asymptotics of partial sums can be explored. Examining;

$$\alpha(N) = \sum_{n=2}^{N} \frac{1}{\sigma(n)} \quad \beta(N) = \sum_{n=2}^{N} \frac{1}{n\sigma(n)}$$

Equation (4) gives the immediate:

$$0 < \sum_{n=2}^{N} \frac{1}{\sigma(n)} - P_N(0) \leq \sum_{n=2}^{N} \frac{1}{n^0} \Rightarrow 0 < \sum_{n=2}^{N} \frac{1}{\sigma(n)} - \pi(N) \leq N - 1 \Rightarrow 0 < \frac{1}{N} \sum_{n=2}^{N} \frac{1}{\sigma(n)} - \frac{\pi(N)}{N} \leq \frac{N-1}{N}$$

Which, since the prime-counting-function $\pi(x)$ is obviously less than x, gives the asymptotic equation:

$$\sum_{n=2}^{x} \frac{1}{\sigma(n)} \sim cx \quad 0 < c \leq 1$$

Similarly,

$$0 < \sum_{n=2}^{N} \frac{1}{n\sigma(n)} - P_N(1) \leq \sum_{n=2}^{N} \frac{1}{n}$$

Using Merten's Theorem (or simply the trivial fact that $P_N(1) < H_N$) and standard Harmonic number approximations:

$$0 < \frac{1}{\ln N} \sum_{n=2}^{N} \frac{1}{n\sigma(n)} - \frac{(\ln\ln N + M)}{\ln N} \sim \frac{\gamma + \ln N - 1}{\ln N} \Rightarrow \sum_{n=2}^{x} \frac{1}{n\sigma(n)} \sim c \ln x \quad 0 < c \leq 1$$

A numerical table of these asymptotics is listed below. Computation courtesy of Anjay Singla.

Upper Bound = N	$\beta(N)$	$\beta(N)/ln(N)$	Small N Approx:$\beta(N)/(e^{\gamma} lnln(N))$	$\alpha(N)$	$\alpha(N)/N$
10^3	3.31717	0.4802	0.9637	338.2307	0.3382
10^4	3.9626	0.4302	1.002	2789.3439	0.2789
10^5	4.5205	0.3926	1.0387	24111.645	0.2411
10^6	5.01719	0.3632	1.0728	214685.2742	0.2147
10^7	5.4682	0.3393	1.1044	1949922.4336	0.195

More generally of course, this reciprocal-divisor methodology applies to all sums of the form;

$$f_N(s) = \sum_{n=1 \ or \ 2}^{N} g(n)\, \hat{p}(n)$$

Or if one works with twin primes, $\hat{p}(n)\hat{p}(n+2)$.

4 Inverting a Counting Function: Filtering Out The Nth Term

To find the Nth member of a set, such as the primes for example, one poses the two following questions: When does the counting function equal N? *and* Is N a member of the set?

For the twin primes as example, that is to say when does $2\tau(x) - 2N + 1$ equal 1 (any integer factor, here 2, works to ensure the expression is nonzero). In place of the floor function which is widely used for this purpose, $\hat{D}(1, x)$ acts as an indicator for if $x = 1$.

Finally by filtering with these conditions over all possible n, we achieve the desired result. For a strictly increasing counting function $C(x)$, the Nth counted member is equal to:

$$c_N = \sum_{n=1}^{\infty} n \cdot \hat{D}(1, 2C(n) - 2N + 1) \cdot [C(n) - C(n-1)] \tag{5}$$

The famous Willian's formula for primes achieves a similar 'inversion' result with *much* greater compactness using the floor function as a logic operator of sorts.

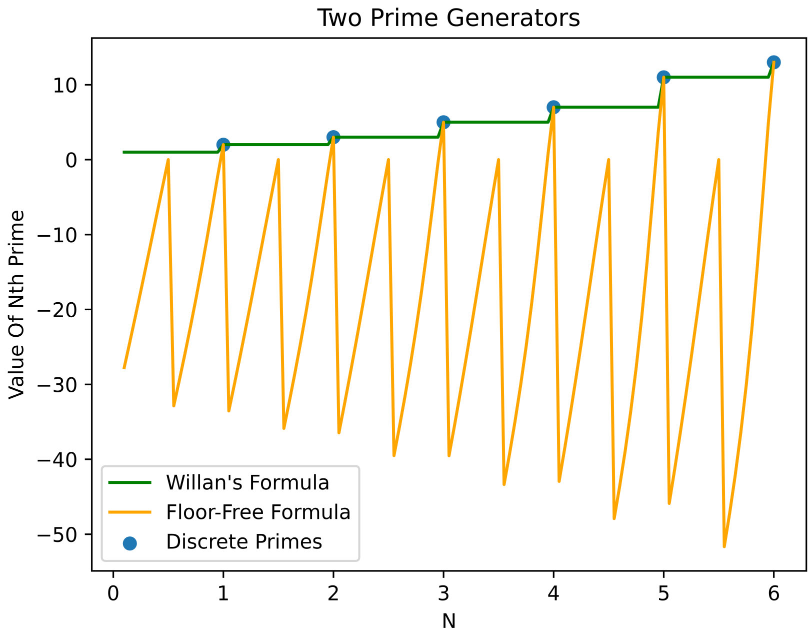
$$c_N = 1 + \sum_{n=1}^{\infty} \lfloor \left(\frac{N}{C(n) + 1} \right)^{\frac{1}{N}} \rfloor$$

Inserting the many-times-nested definitions from Section 1 soon shows the patent trade-off of Equation 5 - In exchange for the novelty of no floor brackets, summations are nested together in highly non ideal ways. Following (5), the nth prime, p_n and the nth twin prime t_n may then be expressed as:

$$p_N = \sum_{n=2}^{\infty \text{ or } 2^N} n \cdot \hat{D}\left(1, 2\left(\sum_{i=1}^{n} \hat{p}(i) - N + \frac{1}{2}\right)\right) \cdot \hat{D}\left(2, \sum_{j=1}^{n} \hat{D}(n, j)\right) \tag{6}$$

$$t_N = \sum_{n=2}^{\infty} n \cdot \hat{D}\left(1, 2\left(\sum_{i=1}^{n} \hat{p}(i)\, \hat{p}(i+2) - N + \frac{1}{2}\right)\right) \cdot \hat{D}\left(2, \sum_{j=1}^{n} \hat{D}(n, j)\right) \hat{D}\left(2, \sum_{j=1}^{n+2} \hat{D}(n+2, j)\right) \tag{7}$$

Where $\hat{D}\left(2, \sum_{j=1}^{n} \hat{D}(n,j)\right) = \hat{p}(n)$ from Section 1. Continuing with the impractical novelties, by omitting floor functions, the above formulas smoothly interpolate the primes, allowing for the existence of *fractional* primes, as graphed below. The aforesaid Willian's formula is graphed as well. Summation bounds [2,99] with x-increments of 0.05



5 Generalizations

Taking a further inspection of the fundamental equation (1), a relation with the DFT is immediately evident. Beginning with the Inverse DFT definition for vector x with N terms; (Subscript denotes the element at a position within a vector)

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} \left(DFT\left\{x\right\}\right)_k e^{\frac{2\pi i}{N} kn}$$

In fact, the cyclic property of the DFT with period N is precisely the mechanism needed. Consider;

$$x_{n+N} = \frac{1}{N} \sum_{k=0}^{N-1} \left(DFT\left\{x\right\}\right)_k e^{\frac{2\pi i}{N} k(n+N)} = \frac{1}{N} \sum_{k=0}^{N-1} \left(DFT\left\{x\right\}\right)_k e^{\frac{2\pi i}{N} kn} \cdot 1 = x_n$$

To illustrate this, n dividing N represents a cycle of length N starting with 1 and followed by 0. When n cycles cleanly back to the beginning of vector x, x_n will be 1, and 0 otherwise - producing the divisor indicator (1).

$$[1,0,0,...]_n = \frac{1}{N} \sum_{k=0}^{N-1} \left(DFT\left\{[1,0,0,...]\right\}\right)_k e^{\frac{2\pi i}{N} kn} = \frac{1}{N} \sum_{k=0}^{N-1} 1 \cdot e^{\frac{2\pi i}{N} kn} \Rightarrow \frac{1}{N} \sum_{k=0}^{N-1} \cos\left(\frac{2\pi}{N} kn\right)$$

This methodology is intriguing in that it performs a cycling of *any* vector x over any length N. Any discrete cyclic pattern can be modeled by this technique, as I will illustrate through the simple modulo operator.

$$n \bmod N = [0,1,2...,N-1]_n = \frac{1}{N} \sum_{k=0}^{N-1} \left(DFT\left\{[0,1,2...,N-1]\right\}\right)_k e^{\frac{2\pi i}{N} kn}$$

Computing the necessary DFT matrix follows algebraically.

$$\left(DFT\left\{[0,1,2...,N-1]\right\}\right)_k = \sum_{m=0}^{N-1} m e^{\frac{-2\pi i k m}{N}}$$

The first term ($k=0$) will be $\sum_{m=0}^{N-1} m = \frac{N(N-1)}{2}$, and from here, a differentiated geometric series may be used. Since,

$$\sum_{m=0}^{N-1} m x^m = \frac{(N-1)\,x^{(N+1)} - Nx^N + x}{(x-1)^2}$$

$$\sum_{m=0}^{N-1} m\left(e^{\frac{-2\pi i k}{N}}\right)^m = \frac{N}{e^{\frac{-2\pi i k}{N}} - 1} = \frac{N}{2}\left(-1 + i\frac{\sin\left(\frac{2\pi k}{N}\right)}{1-\cos\left(\frac{2\pi k}{N}\right)}\right)$$

Putting everything together, one obtains:

$$A \bmod n = \frac{1}{n}\left(\frac{n(n-1)}{2} + \sum_{k=1}^{n-1}\frac{n}{2}\left(-1 + i\frac{\sin\left(\frac{2\pi k}{n}\right)}{1-\cos\left(\frac{2\pi k}{n}\right)}\right)e^{-\frac{2\pi i k A}{n}}\right)$$

$$A \bmod n = \left(\frac{(n-1)}{2} - \frac{1}{2}\sum_{k=1}^{n-1}\cos\left(\frac{2\pi k A}{n}\right)\right) - \frac{1}{2}\sum_{k=1}^{n-1}\sin\left(\frac{2\pi k A}{n}\right)\cot\left(\frac{\pi k}{n}\right) \tag{8}$$

Where if $n \nmid A$, the divisor indicator expression (1) reduces to;

$$\frac{n}{2} - \frac{1}{2}\sum_{k=1}^{n-1}\sin\left(\frac{2\pi k A}{n}\right)\cot\left(\frac{\pi k}{n}\right) \quad n \nmid A$$